Progress Report 2: Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | October 13, 2026

1. Tasks Completed

Task	What was built / implemented	Hours	Results and outcomes
Dataset access investigation	Checked how NewsCLIPpings and VisualNews can be obtained and documented each blocker in a blocker log.	1.5	VisualNews is now an open download (no request form) but ships as one 91 GB archive. Dead image URLs are not an issue: annotations use local VisualNews ids.
Data download, validation, curated subset	Download script and data-prep script: fetched NewsCLIPpings (annotations, metadata, CLIP ViT-B/32 embeddings, 2.6 GB), validated it, and built a stratified subset.	2.0	All splits exactly 50/50 genuine vs. falsified, 0 duplicates, 0 missing values, 100% embedding coverage. Curated subset: 24,000 pairs (19,200 / 2,400 / 2,400), balanced across all 4 mismatch methods.
Embedding-generation experiment	Script that streams part of the VisualNews archive, maps images to NewsCLIPpings ids, and generates my own CLIP embeddings on a Mac GPU (MPS).	2.0	2,641 images matched. Agreement with the provided embeddings: mean cosine 0.994 (images), 0.999999 (captions). Speed: 224 images/s end to end, so Colab is not needed.
Similarity-threshold baseline	Cosine-similarity classifier with the threshold tuned on validation; test metrics with bootstrap 95% CIs.	1.5	Test ROC-AUC 0.716 (CI 0.695 to 0.734), accuracy 0.637, F1 0.580. By method: CLIP text-text 0.910, scene 0.744, person 0.705, CLIP text-image 0.458 (below chance).
Per-method visualizations	Similarity histograms, ROC curves, per-method AUC/F1 chart, PCA and t-SNE plots of 2,000 test pairs.	1.0	Genuine and falsified pairs do not separate in 2-D (AUC 0.46 to 0.51 on 2-D coordinates) but cluster clearly by method, so the signal needs the full 512-dim embeddings.
Documentation and repository	Updated the Design Document (v2), wrote this report, organized the project folder and set up the GitHub repository with a README.	1.5	Design Document timeline now shows Weeks 3 to 6 complete; subset-size and dead-URL decisions resolved.
Total		**9.5**	

Blockers, challenges, and risks

- **Cloud environment blocked the dataset hosts** (Hugging Face and the VisualNews server returned 403). Solved by running everything on my own laptop.
- **VisualNews downloads at about 1.7 MB/s** (about 15 hours for 91 GB). Solved with HTTP range requests: I streamed only the first 1.5 GB, and modeling uses the provided embeddings, so it does not need every image.
- **A partial archive ends in a read error.** The extraction script stops cleanly at the cut-off and skips the truncated file.
- **Risk:** the CLIP text-image split was built from pairs CLIP already rates as similar, so the baseline scores below chance there. The trained classifiers need to close this gap.

2. Plan for Next Week

- Train a logistic regression classifier on the full CLIP embeddings (image, caption, their difference and element-wise product); tune regularization on validation.
- Compare it to the baseline overall and per method with bootstrap 95% CIs, and check whether it beats chance on the CLIP text-image split.
- Find the cause of the small image-embedding differences (likely resize/crop preprocessing) so the future Streamlit app matches the training embeddings.
- Draft the lightweight neural network architecture for Week 9.

Help needed

None at this time. Two choices guide next week's work:

- Classifiers are trained on NewsCLIPpings' provided CLIP embeddings, since my own pipeline reproduces them (mean cosine 0.994), so modeling does not wait on downloading 91 GB of images.
- The CLIP text-image split is reported alongside the other methods and also treated as a stress test, since it was designed to defeat CLIP similarity.